



COURSE SYLLABUS						
Cours e Code	COMP 004	Course Title	Discrete Structures 1		Course Credit	3
Course Description		This course provides the essential mathematical toolkit for modern Information Technology. The focus is on applying formal reasoning to solve practical IT problems, designing efficient systems, and validating their correctness. This globally competitive curriculum is designed to equip students with the rigorous analytical skills sought by employers in the technology industry, preparing students to design, secure, and manage the complex systems that drive national and global development.				
Pre-Requisites		GEED 004		Co-Requisites	None	
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				

2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia

The WORLD UNIVERSITY RANKINGS for INNOVATION

Rank 21
GENERATIVE AI APPLICATION

Rank 39
STUDENT SUPPORT ENGAGEMENT

Rank 50
SYMBOL PROMOTION

5 STARS

Employability

Online Learning

Internationalization

Teaching

Academic Development

Social Responsibility

Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)

Times Higher Education
Impact Rankings 2024
DEBUTS AT 1501+



**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





	6. Leadership and Organizational Skills. Graduates assume leadership roles and become leading professionals in their respective disciplines by equipping them with appropriate organizational skills. 7. Personal and Professional Ethics. Graduates manifest integrity and adherence to moral and ethical principles in their personal and professional circumstances. 8. Resilience and Agility. Graduates demonstrate flexibility and the growth mindset to adapt and thrive in the volatile, uncertain, complex and ambiguous (VUCA) environment. 9. National and Global Responsiveness. Graduates exhibit a deep sense of nationalism as it complements the need to live as part of the global community where diversity is respected. They promote and fulfill various advocacies for human and social development.
COLLEGE GOALS AND OBJECTIVES	The Polytechnic University of the Philippines – Taguig Campus aims to achieve the following goals: 1. Offer curricula which are relevant and responsive to the changing needs of the industry and society; 2. Provide trainings to students that will enable them to become potent instruments to socio-economic development. 3. Prepare students to become productive, creative, innovative, and dynamic in their chosen fields of specialization. 4. Provide state-of-the-arts facilities of learning to optimize student development; 5. Inculcate appropriate values that are necessary to build a humane, disciplined, nationalist, and independent society; 6. Tap potentials of students, faculty, administrative staff and other stakeholders in formulating policies for institutional development; 7. Involve all stakeholders in enhancing programs for student’s faculty and employees’ development; 8. Partner with industry and other research institutions strengthening research capabilities of faculty, employees and students; 9. Facilitate presentation of research outputs in international fora, and their publication in recognized journals; 10.Undertake outreach and research-based extension programs by tapping the expertise and other resources of all stakeholders and, 11.Produce technologies for commercialization of livelihood improvement.

PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	QS WORLD UNIVERSITY RANKINGS 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia WURI The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL/ PROMOTION QS STARS ★★★★★ Employability ★★★★★ Teaching ★★★★★ Academic Development ★★★★★ Social Responsibility Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND) ★★★★★ Internationalization Times Higher Education Impact Rankings 2024 DEBUTS AT 1501+ o ILOs <table><tr><td>5</td><td>7</td><td>8</td><td>9</td></tr><tr><td></td><td></td><td></td><td></td></tr></table>	5	7	8	9				
	5	7	8	9						
Apply knowledge of computing, science, and mathematics appropriate to th										



	Analyze complex problems, and identify and define the computing requirements appropriate to its solution											
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.											
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.											

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	Integrate IT-based solutions into the user environment effectively.									
	Assist in the creation of an effective IT project plan.									
	Apply knowledge through the use of current techniques, skills, tools and practices necessary for the IT profession									
	Understand best practices and standards and their applications.									
	Function effectively as a member or leader of a development team recognizing the different roles within a team to accomplish a common goal.									
	Communicate effectively with the computing community and with society at large about complex computing activities through logical writing, presentations, and clear instructions.									
	Analyze the local and global impact of computing information technology on individuals, organizations, and society.									
	Understand professional, ethical, legal, security and social issues and responsibilities in the utilization of information technology.									
	Recognize the need for and engage in planning self-learning and improving performance as a foundation for continuing professional development									

COURSE LEARNING OUTCOMES (CLO)	1. Explain fundamental concepts of logic, sets, relations, and functions relevant to computer science and information technology.	Alignment to PLOs							
	2. Apply rules of propositional and predicate logic to analyze and construct proofs.	1	2	3	4	5	6	7	8
	3. Perform operations on sets, relations, and functions to model and solve c								

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	4. Illustrate various counting techniques, including permutations and combinations, to solve discrete probability and combinatorial problems.								
	5. Use mathematical induction and recursion to define and analyze sequences and algorithms								
	6. Represent and analyze discrete structures such as graphs and trees in solving problems related to networks and data organization.								
	7. Demonstrate problem-solving and critical thinking skills using discrete mathematical concepts in algorithmic contexts.								
	8. Integrate discrete mathematical reasoning in the design and analysis of computer programs and systems.								

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OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	- Equip students on the pivotal ideas embedded in the vision, mission, and 10 Pillars of PUP - Gain an overview of the course subject and requirements	- Orientation about PUP Vision, Mission, and 10 Pillars - Overview about the Subject Course and Course Requirements	Present the VMGO, policies, and guidelines on the conduct of Face-to-Face; and of the requirements of the course.		Formative Assessment (Sharing of Learning)		X			
2	Define basic logical concepts and identify propositions and logical connectives.	Introduction: mathematical logic, statements, logical connectives, truth tables	Lecture + interactive logic puzzles, small group: evaluate compound statements	Assigned reading, online logic tutorial videos	Quiz 1 (truth tables, basic connectives)					
3	Demonstrate the use of logical equivalences and laws to simplify logical expressions.	Logical equivalences, laws (De Morgan, distributive, etc.), implication, contrapositive, biconditional	In-class proof of equivalences, student presentations	Live explanation, breakout for proof work	Homework 1					
4	Translate verbal statements into	Predicate logic: predicates,	Class translation exercises ("for all", "there exists" statements)	Live peer review of student pro						



	symbolic logic and vice versa using predicates and quantifiers.	quantifiers, domain, scope, nested quantifiers			negate quantified statements					
5	Apply direct, contrapositive, and contradiction proofs in problem solving.	Methods of proof: direct, contrapositive, contradiction, counterexample	In-class proof writing workshops	Recorded lectures, extra proof sets	Homework 2: write short proofs					
6	Perform basic set operations and visualize them using Venn diagrams.	Basic set theory: sets, subsets, power sets, operations (union, intersection, complement)	Hands-on Venn diagram exercises, set operations	Reading and online exercises	Quiz 3: set operations, Venn diagrams					
7	Apply advanced set operations such as Cartesian product and inclusion-exclusion principle.	Advanced set operations: Cartesian product, indexed families, set algebra, inclusion-exclusion principle	In-class examples, student mini presentations	Live walkthrough of inclusion-exclusion	Homework 3: inclusion-exclusion problems					





8	- Identify and classify relations based on their properties and define equivalence relations. - Compute relation operations and closures to model relationship.	- Relations: definitions, properties (reflexive, symmetric, transitive), equivalence relations and partitions - Operations on relations: composition, inverses, closures (transitive, reflexive closures)	Workshop: examine relation examples, equivalence classes In-class closure computation, small group tasks	Live group exercises Lecture recording, closure algorithm examples	Homework 4: compute closures and composition					
9	MIDTERM WRITTEN EXAMINATION									
10	Analyze and construct functions; determine injectivity, surjectivity, and bijectivity.	Functions: definition, injective, surjective, bijective, inverse functions, composition	In-class mapping exercises, student-led proofs of bijection	Live concept check, breakout for proofs	Quiz 4: sum/product and pigeonhole					
11	Apply counting rules to compute possible outcomes in structured	Basic counting: rule of sum, rule of product, pigeonhole principle	In-class counting puzzles, brainstorming solutions	Recorded lectures and problem sets	 2025 7th in the Philippines 92nd in Southeast Asia 541-560 in Asia  WURI The WORLD UNIVERSITY RANKINGS for INNOVATION Rank 21 GENERATIVE AI APPLICATION Rank 39 STUDENT SUPPORT ENGAGEMENT Rank 50 SYMBOL/ PROMOTION  QS STARS Employability Online Learning Internationalization Teaching Academic Development Inclusiveness Social Responsibility Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BSND)  Times Higher Education Impact Rankings 2024 DEBUTS AT 1501+ <td></td> <td></td> <td></td> <td></td> <td></td>					



	problems.									
12	Solve problems involving permutations and combinations.	Permutations & combinations, combinations with repetition, binomial coefficients, Pascal's identity	Group work: combinatorial tasks, online combinatorics tool	Asynchronous combinatorics problems	Homework 5: permutations/combinations sets					
13	Employ inclusion-exclusion and basic generating functions to solve combinatorial problems.	Advanced counting: inclusion-exclusion in combinatorial problems, generating functions	In-class combinatorial problem solving	Recorded module and extra practice	Quiz 6: inclusion-exclusion, simple generating functions					
14	Use mathematical induction to prove propositions and formulas.	Mathematical induction: principle, strong induction, well-ordering	Proof workshop, students present induction proofs	Video tutorials and proof exercises	Homework 6: induction proofs (sum formulas, inequalities)					
15	Formulate and solve recurrence relations in algorithmic contexts.	Recurrence relations, homogeneous/non-homogeneous, solving recurrences, recurrence in algorithms	In-class derivation of recurrence solutions, algorithmic examples	Live solving recurrence examples	Quiz 7: recurrence relation solving					
16	Describe and represent basic graph structures and their	Graph theory basics: definitions, terminology (vertices, edges, degree, path, cycle, connectedness)	In-class graph drawing, student small group tasks	Asynchronous graph theory readings, vide						

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QS STARS
★★★★★
- Employability
- Online Learning
- Internationalization
★★★★★
- Teaching
- Academic Development
- Inclusiveness
★★★★★
- Social Responsibility
- Specialist Criteria: Bachelor of Science in Nutrition and Dietetics (BEND)

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	properties.										
17	Apply graph traversal and spanning tree algorithms to solve network problems.	Trees, tree traversals, application in data organization; advanced graph problems	In-class algorithm walkthroughs, coding pseudocode Workshop: tree traversal, representation, network modeling	Recorded algorithm walk-throughs and coding tasks Asynchronous tree traversal videos, problem sets	Quiz 8: algorithm concepts, pseudocode understanding						
18	FINAL WRITTEN EXAMINATION										

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		Resources and Applying Strategies in Teaching Prose <i>Competency 4:</i> Recognizing Assessment in Teaching Prose	achievement which are consistent with the selected competencies • Learning Plan Making according to the English curricula that develops higher order thinking skills of learners through the use of prose and dramatic texts		• Quizzes					
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FINAL EXAMINATION

LIST OF REFERENCES

BOOKS:

WEBSITE:

CalcWorkshop. (2023, April 1). Learn discrete math (with videos & plenty of practice!). Retrieved from <https://calcworkshop.com/discrete-math/>

Coursera. (n.d.). Introduction to logic. Retrieved 2025, from <https://www.coursera.org/learn/logic-introduction>

Coursera. (n.d.). Introduction to discrete mathematics for computer science specialization. Retrieved 2025, from <https://www.coursera.org/specializations/discrete-mathematics>

Logic and Proof: 2024-2025. (n.d.). University of Oxford. Retrieved 2025, from <https://www.cs.ox.ac.uk/teaching/courses/2024-2025/logicandproof/>

Discrete Mathematics: 2023-2024. (n.d.). University of Oxford. Retrieved 2025, from <https://www.cs.ox.ac.uk/teaching/courses/2023-2024/discretemaths/>

Open Mathematics — Discrete Mathematics: An Open Introduction (3rd ed.). (n.d.). Retrieved 2025, from <https://discrete.openmathbooks.org/dmoi3.html>

Brooklyn College / CUNY. (2025, April 28). CISC 2210 Introduction to Discrete Structures (Briskman) – LibGuides. Retrieved from <https://libguides.brooklyn.cuny.edu/cisc2210-briskman>](<https://libguides.brooklyn.cuny.edu/cisc2210-briskman>)

LogicLearner: A tool for guided practice of propositional logic proofs. (2025, March 25). *arXiv*. Re

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CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students' Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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5+ stars

Employability
Online Learning
Internationalization



4+ stars

Teaching
Academic
Development
Inclusiveness



3+ stars

Social Responsibility
Specialist Criteria: Bachelor
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Bachelor of Science in Information Technology
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Website: www.pup.edu.ph | Inquiries: <https://bit.ly/PUPSINTA>

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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.

We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.

Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.

Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.

Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.

Class announcements will be communicated through the class page or posted through university-issued (i.e., official) email accounts. Students should check their email regularly.

Intellectual Property and Copyright.

Original materials created for this online class is covered by copyright. If you use any (third-party) instructional materials (e.g., YouTube videos) might be subject to copyright while others (e.g., some assigned readings) might be subject to fair use.



Always be honest.

Some materials will be available for download.



be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a





Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	30%	Final Examinations	
	100%		30%
			100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
Performance Assessments and Written Exercises Students must complete weekly quizzes, homework assignments, and written problem sets covering logic, set theory, relations, counting, induction, and graph theory. Major Examinations Two examinations will be conducted — Midterm (Week 9) and Final Exam (Week 18) — to evaluate mastery of theoretical	Final Grade = (Midterm Grade + Final Term Grade) / 2		





and applied discrete mathematics concepts.

Project and Presentation

Students will design and present a mini research or algorithmic project (Weeks 17–18) applying discrete structures such as trees, graphs, or recurrence relations to real-world computing problems.

CLASS INFORMATION	FACULTY INFORMATION
Course, Year and Section: BSIT 1-1	Name of Faculty: Villacorta, Karl Kristian Gonzaga
Day and Time: Tuesday, 10:30AM – 01:30PM	Consultation Time:
Room: TGA 402	Office Tel. No./ Mobile Phone No.
School Year: 2025-2026 Semester: 2 nd	E-Mail Address:

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Karl Kristian G. Villacorta Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES mpuses

